

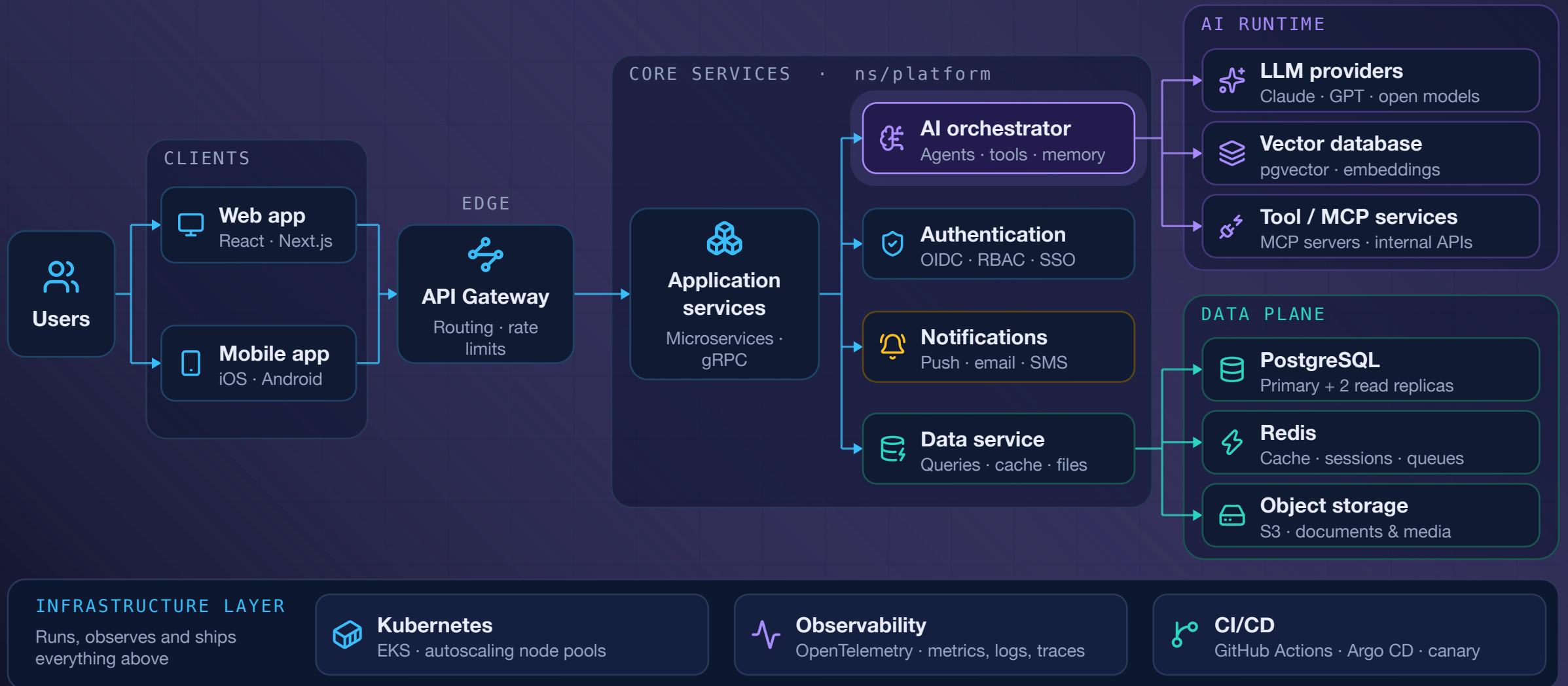
AI Platform — Production Architecture

ARCHITECTURE REVIEW

rev 2.3 · prod / us-east-1

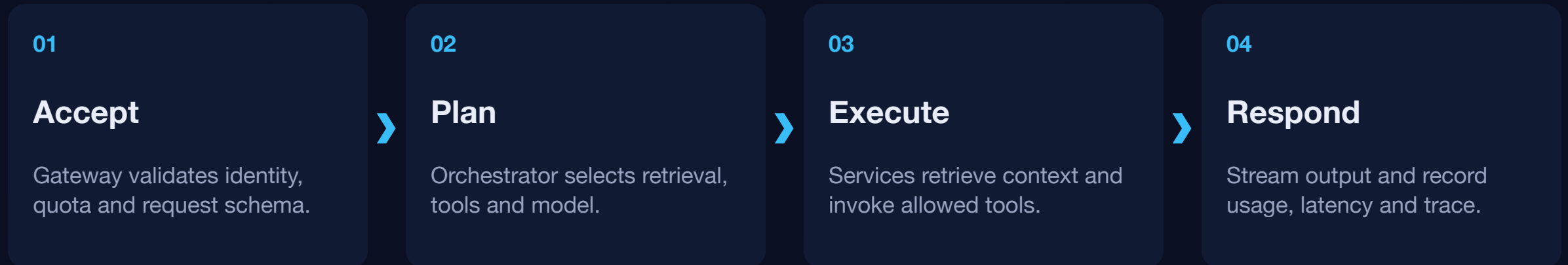
Scalable, secure and modular architecture for AI-powered applications

● Request path ● AI runtime ● Data plane



One request. Four deliberate stages.

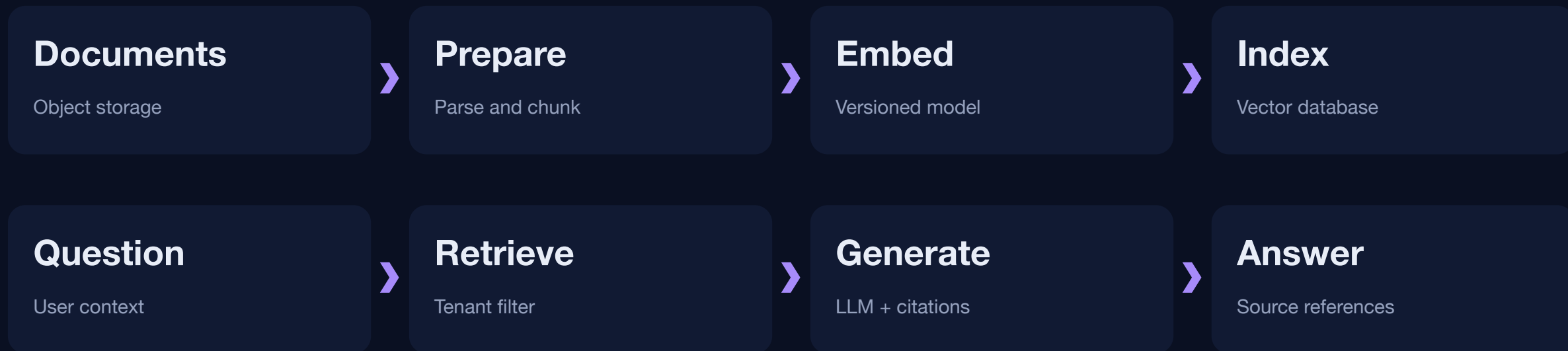
Bounded retries, streaming responses and trace context make the path production-ready.



One trace ID spans the gateway, orchestrator, tool calls and model response.

Ground the model in governed knowledge.

Separate ingestion from retrieval; keep citations and access control attached to context.



Enforce document permissions during retrieval; cache only within a tenant boundary.

Trust boundaries are part of the architecture.

Identity, policy and tool permissions are enforced before an action reaches a system.



Identity

OIDC and short-lived tokens
Tenant-scoped service calls
Least-privilege credentials



Execution policy

Tool allowlists and budgets
Human approval for writes
Sandboxed execution

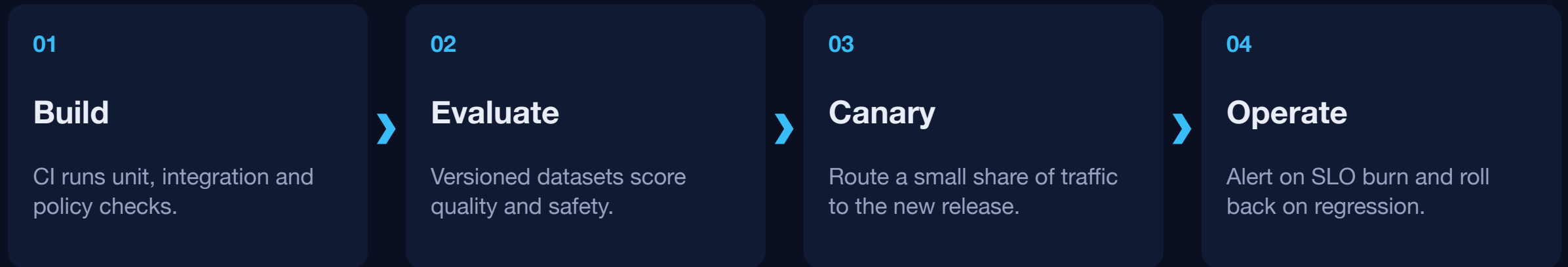


Audit and data

Encrypted storage and transport
Redacted operational logs
Immutable action history

Ship safely. Observe continuously.

Illustrative service targets: 99.9% availability and p95 first token below 1.5 seconds.



Kubernetes runs the platform; observability connects release versions to live behavior.